CECH COHOMOLOGY IN DEGREE ONE AGREES WITH COHOMOLOGY ON EVERY TOPOLOGICAL SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and concludes surjectivity through the vanishing of the sheafification; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-cech-always is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- The Cech-to-cohomology spectral sequence and its low degree terms. Same chapter, Lemma lemma-cech-spectral-sequence, invoked at proof lines 2651-2652. Supplies the exact sequence from which injectivity and the surjectivity reduction are read.
- Locality of cohomology: every cohomology class dies on a covering. Same chapter, Lemma lemma-kill-cohomology-class-on-covering, invoked at proof lines 2662-2663. Supplies that the sheafification of the presheaf of first cohomology vanishes.
- The comparison map from Cech cohomology to cohomology. Same chapter, Equation equation-cech-to-cohomology, named in the statement at line 2646. Supplies the map whose bijectivity is the content of the lemma.

Source reference. Original equation 09UZ: sources/stacks-cohomology/pinned-original-source.tex, source lines 2511–2511. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01ES: sources/stacks-cohomology/pinned-original-source.tex, source lines 1541–1541. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01E3: sources/stacks-cohomology/pinned-original-source.tex, source lines 559–559. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a topological space. Let $\mathcal{F}$ be an abelian sheaf. Then the map $\check{H}^1(X, \mathcal{F}) \to H^1(X, \mathcal{F})$ defined in (09UZ) is an isomorphism.*

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Proof. Let $\mathcal{U}$ be an open covering of X. By Lemma 01ES there is an exact sequence

$$0 \to \check{H}^1(\mathcal{U}, \mathcal{F}) \to H^1(X, \mathcal{F}) \to \check{H}^0(\mathcal{U}, \underline{H}^1(\mathcal{F}))$$

Thus the map is injective. To show surjectivity it suffices to show that any element of $\check{H}^0(\mathcal{U}, \underline{H}^1(\mathcal{F}))$ maps to zero after replacing $\mathcal{U}$ by a refinement. This is immediate from the definitions and the fact that $\underline{H}^1(\mathcal{F})$ is a presheaf of abelian groups whose sheafification is zero by locality of cohomology, see Lemma 01E3. $\square$